Date: 02 July 2026	**Team ID:** SB-7429
Project Name: Credit Card Approval Prediction using Machine Learning	**Maximum Marks:** 5 Marks

CODE-LAYOUT, READABILITY AND REUSABILITY TEMPLATE

Code Quality Parameter Checklist

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Uniform 4-space indentations conforming to PEP 8 styles.	Yes	Checked via linter.
2	Proper File Structure	Decoupled structures (app.py, model.py, static/, templates/).	Yes	Highly modular.
3	Meaningful Variables	Clear names reflecting attributes (e.g. AMT_INCOME_TOTAL).	Yes	Readable code.
4	Function Names	Verbs defining processes (e.g. apply_policy_rules()).	Yes	Self-documenting.
5	Code Comments	Inline comments detailing SMOTE, scaling, and overrides.	Yes	Comprehensive.
6	Modular Design	Isolated preprocessing, model pipeline, and web routing.	Yes	Clean separations.
7	No Redundant Code	Reusable functions for variable engineering and scaling.	Yes	DRY compliant.
8	Error Handling	Try-except blocks handle missing form values and file operations.	Yes	Robust fallback.

Reusable Components / Modules

S.No	Component / Module Name	Language	Where Reused	Reusability Level (High/Medium/Low)
1	engineer_features()	Python	Used in model.py training and app.py real-time inference.	High
2	StandardScaler (scaler.pkl)	Python / Pickle	Fitted during training, loaded for prediction feature scaling.	High
3	apply_policy_rules()	Python	Used in Flask predictive router and automated testing suites.	High

Overall Code Quality Assessment

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Highly organized folder structures separating data, logic, and views.
Readability & Comments	5	Self-documenting variable names with detailed comments explaining calculations.
Modularity & Reusability	5	Features engineered via shared functions across environments.